

Appendix A: Results of Experiment 1

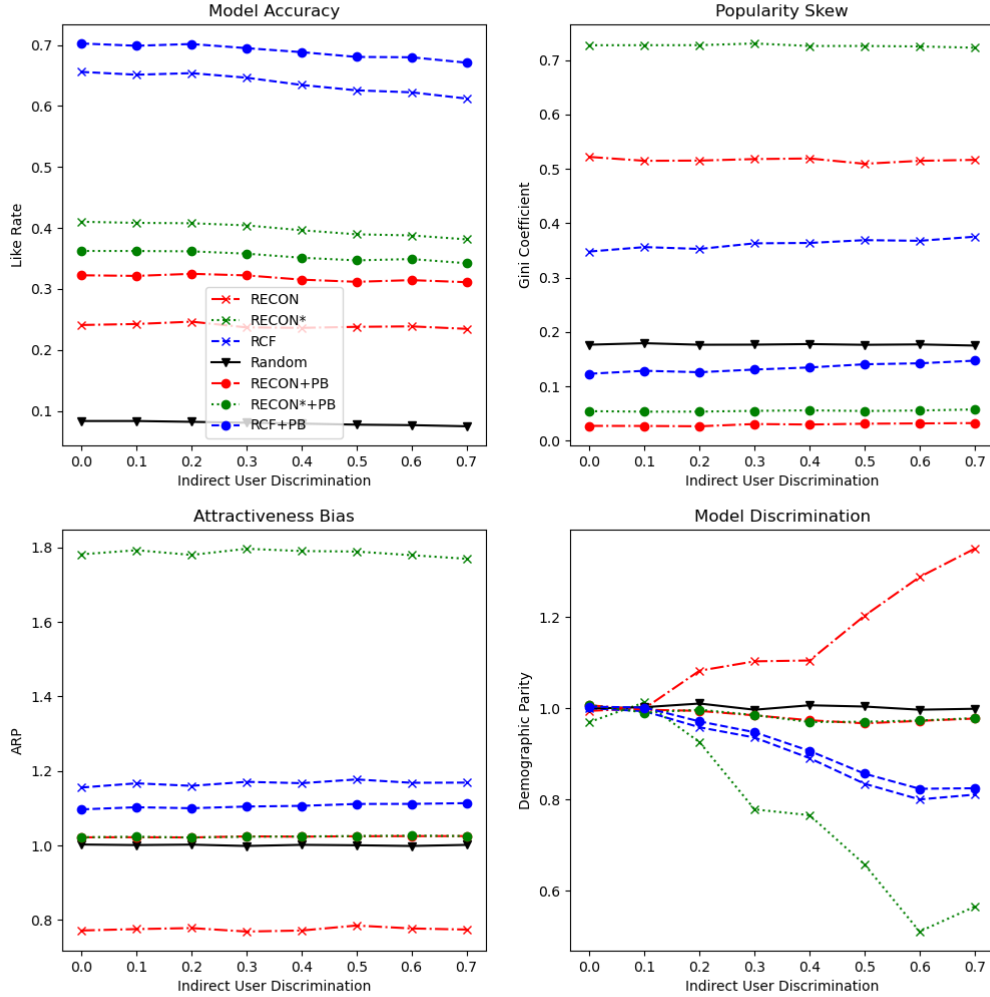


Fig. 1. The results of indirect user discrimination on model performance. Left top (1a): model accuracy, as measured by like rate. Right top (1b): Popularity skew, as measured by Gini Coefficient. Left bottom (1c): Attractiveness Bias, as measured by ARP. Right bottom (1d): Model discrimination, as measured by demographic parity ratio.

Appendix B: Results of Experiment 2

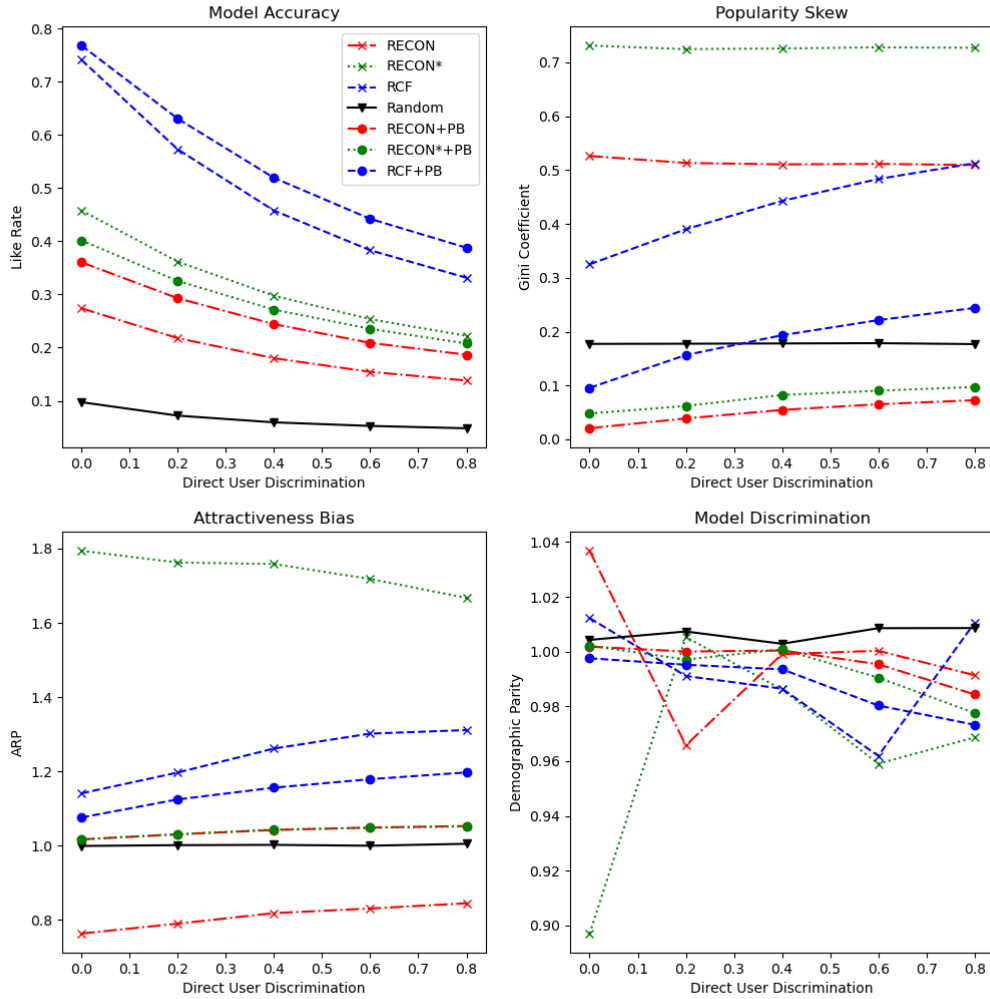


Fig. 2. The results of direct user discrimination on model performance. Left top (1a): model accuracy, as measured by like rate. Right top (1b): Popularity skew, as measured by Gini Coefficient. Left bottom (1c): Attractiveness Bias, as measured by ARP. Right bottom (1d): Model discrimination, as measured by demographic parity ratio.

Appendix C: Full scores of Experiment 1

Recommender	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.5$	$\lambda = 0.6$	$\lambda = 0.7$
RECON	0.241 (0.01)	0.243 (0.009)	0.246 (0.011)	0.237 (0.01)	0.236 (0.008)	0.238 (0.008)	0.239 (0.012)	0.235 (0.011)
RECON+PB	0.322 (0.011)	0.321 (0.012)	0.325 (0.011)	0.322 (0.008)	0.315 (0.008)	0.312 (0.012)	0.315 (0.013)	0.311 (0.01)
RECON*	0.41 (0.016)	0.408 (0.014)	0.408 (0.011)	0.404 (0.012)	0.396 (0.013)	0.39 (0.014)	0.388 (0.015)	0.381 (0.012)
RECON*+PB	0.362 (0.014)	0.362 (0.011)	0.362 (0.01)	0.358 (0.011)	0.351 (0.009)	0.347 (0.013)	0.349 (0.014)	0.342 (0.012)
RCF	0.656 (0.013)	0.651 (0.013)	0.654 (0.017)	0.646 (0.012)	0.634 (0.016)	0.626 (0.018)	0.622 (0.016)	0.612 (0.018)
RCF+PB	0.702 (0.014)	0.699 (0.013)	0.701 (0.012)	0.695 (0.012)	0.688 (0.011)	0.68 (0.014)	0.68 (0.012)	0.671 (0.014)
Random	0.084 (0.003)	0.084 (0.004)	0.082 (0.004)	0.081 (0.004)	0.079 (0.005)	0.078 (0.003)	0.077 (0.004)	0.075 (0.004)

Table 3. Like rate mean and SD for indirect discrimination

Recommender	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.5$	$\lambda = 0.6$	$\lambda = 0.7$
RECON	0.522 (0.012)	0.515 (0.012)	0.515 (0.013)	0.518 (0.011)	0.519 (0.011)	0.509 (0.009)	0.515 (0.01)	0.517 (0.012)
RECON+PB	0.027 (0.005)	0.027 (0.006)	0.027 (0.005)	0.031 (0.006)	0.03 (0.005)	0.031 (0.005)	0.032 (0.004)	0.032 (0.006)
RECON*	0.727 (0.006)	0.727 (0.01)	0.727 (0.008)	0.731 (0.01)	0.726 (0.008)	0.726 (0.007)	0.725 (0.009)	0.723 (0.01)
RECON*+PB	0.054 (0.008)	0.053 (0.009)	0.054 (0.012)	0.055 (0.01)	0.056 (0.009)	0.055 (0.008)	0.056 (0.009)	0.057 (0.007)
RCF	0.348 (0.013)	0.356 (0.014)	0.353 (0.012)	0.363 (0.015)	0.364 (0.014)	0.369 (0.015)	0.368 (0.02)	0.375 (0.018)
RCF+PB	0.123 (0.008)	0.129 (0.01)	0.126 (0.011)	0.131 (0.01)	0.135 (0.01)	0.141 (0.011)	0.142 (0.015)	0.147 (0.014)
Random	0.177 (0.004)	0.179 (0.005)	0.176 (0.004)	0.177 (0.003)	0.178 (0.004)	0.176 (0.005)	0.177 (0.003)	0.175 (0.003)

Table 4. Gini mean and SD for indirect discrimination

Recommender	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.5$	$\lambda = 0.6$	$\lambda = 0.7$
RECON	0.772 (0.02)	0.775 (0.02)	0.778 (0.017)	0.769 (0.021)	0.772 (0.019)	0.785 (0.024)	0.777 (0.023)	0.774 (0.015)
RECON+PB	1.022 (0.004)	1.022 (0.005)	1.021 (0.004)	1.024 (0.005)	1.024 (0.004)	1.025 (0.004)	1.025 (0.004)	1.025 (0.005)
RECON*	1.782 (0.034)	1.793 (0.051)	1.78 (0.04)	1.797 (0.05)	1.791 (0.061)	1.789 (0.041)	1.779 (0.058)	1.77 (0.061)
RECON*+PB	1.022 (0.004)	1.023 (0.005)	1.021 (0.004)	1.023 (0.006)	1.024 (0.005)	1.025 (0.005)	1.026 (0.004)	1.025 (0.006)
RCF	1.155 (0.017)	1.167 (0.015)	1.16 (0.02)	1.171 (0.023)	1.167 (0.019)	1.177 (0.024)	1.168 (0.027)	1.169 (0.023)
RCF+PB	1.097 (0.007)	1.103 (0.01)	1.1 (0.01)	1.104 (0.01)	1.106 (0.009)	1.111 (0.01)	1.112 (0.014)	1.114 (0.012)
Random	1.002 (0.01)	1.001 (0.005)	1.002 (0.006)	0.999 (0.007)	1.002 (0.008)	1.001 (0.006)	0.999 (0.006)	1.002 (0.006)

Table 5. ARP mean and SD for indirect discrimination

Recommender	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.5$	$\lambda = 0.6$	$\lambda = 0.7$
RECON	0.994 (0.133)	1.0 (0.131)	1.082 (0.1)	1.103 (0.125)	1.104 (0.094)	1.202 (0.138)	1.287 (0.12)	1.349 (0.148)
RECON+PB	1.006 (0.019)	0.997 (0.02)	0.994 (0.021)	0.984 (0.024)	0.973 (0.023)	0.967 (0.026)	0.972 (0.033)	0.977 (0.029)
RECON*	0.97 (0.177)	1.013 (0.239)	0.925 (0.19)	0.778 (0.164)	0.765 (0.143)	0.658 (0.151)	0.511 (0.098)	0.565 (0.123)
RECON*+PB	1.007 (0.018)	0.99 (0.024)	0.997 (0.028)	0.985 (0.037)	0.97 (0.03)	0.97 (0.03)	0.973 (0.032)	0.978 (0.035)
RCF	1.0 (0.082)	0.993 (0.095)	0.958 (0.059)	0.936 (0.092)	0.89 (0.106)	0.834 (0.083)	0.8 (0.126)	0.811 (0.102)
RCF+PB	1.003 (0.044)	1.001 (0.033)	0.971 (0.031)	0.947 (0.042)	0.906 (0.053)	0.856 (0.045)	0.823 (0.071)	0.825 (0.062)
Random	0.999 (0.023)	1.002 (0.021)	1.01 (0.039)	0.997 (0.031)	1.006 (0.029)	1.003 (0.033)	0.997 (0.03)	0.999 (0.032)

Table 6. Discrimination mean and SD for indirect discrimination

Appendix D: Full scores of Experiment 2

Recommender	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$
RECON	0.274 (0.011)	0.218 (0.009)	0.18 (0.01)	0.155 (0.007)	0.138 (0.008)
RECON+PB	0.36 (0.012)	0.293 (0.01)	0.244 (0.012)	0.209 (0.008)	0.187 (0.007)
RECON*	0.458 (0.016)	0.362 (0.012)	0.298 (0.013)	0.254 (0.014)	0.222 (0.013)
RECON*+PB	0.402 (0.014)	0.325 (0.011)	0.271 (0.011)	0.235 (0.01)	0.208 (0.009)
RCF	0.741 (0.015)	0.573 (0.016)	0.458 (0.016)	0.383 (0.016)	0.331 (0.014)
RCF+PB	0.768 (0.013)	0.63 (0.014)	0.519 (0.012)	0.442 (0.011)	0.388 (0.012)
Random	0.097 (0.005)	0.072 (0.003)	0.06 (0.003)	0.053 (0.003)	0.048 (0.003)

Table 7. Like rate mean and SD for direct discrimination

Recommender	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$
RECON	0.526 (0.015)	0.513 (0.012)	0.511 (0.01)	0.512 (0.011)	0.509 (0.01)
RECON+PB	0.021 (0.005)	0.039 (0.006)	0.055 (0.009)	0.065 (0.009)	0.073 (0.009)
RECON*	0.731 (0.008)	0.725 (0.009)	0.726 (0.011)	0.728 (0.011)	0.727 (0.008)
RECON*+PB	0.048 (0.01)	0.062 (0.009)	0.082 (0.012)	0.09 (0.011)	0.097 (0.012)
RCF	0.325 (0.014)	0.39 (0.018)	0.443 (0.019)	0.484 (0.02)	0.513 (0.019)
RCF+PB	0.096 (0.009)	0.157 (0.015)	0.193 (0.016)	0.221 (0.018)	0.244 (0.015)
Random	0.177 (0.004)	0.178 (0.004)	0.178 (0.003)	0.179 (0.003)	0.177 (0.003)

Table 8. Gini mean and SD for direct discrimination

Recommender	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$
RECON	0.763 (0.025)	0.79 (0.02)	0.818 (0.024)	0.831 (0.031)	0.845 (0.038)
RECON+PB	1.017 (0.004)	1.031 (0.005)	1.042 (0.008)	1.049 (0.007)	1.053 (0.006)
RECON*	1.794 (0.044)	1.763 (0.066)	1.758 (0.089)	1.718 (0.109)	1.667 (0.099)
RECON*+PB	1.017 (0.006)	1.031 (0.008)	1.043 (0.01)	1.048 (0.011)	1.052 (0.011)
RCF	1.141 (0.017)	1.197 (0.031)	1.262 (0.05)	1.302 (0.062)	1.311 (0.059)
RCF+PB	1.076 (0.008)	1.125 (0.014)	1.156 (0.018)	1.179 (0.018)	1.197 (0.016)
Random	0.999 (0.009)	1.001 (0.008)	1.002 (0.012)	1.0 (0.012)	1.005 (0.013)

Table 9. ARP mean and SD for direct discrimination

Recommender	$\gamma = 0.0$	$\gamma = 0.2$	$\gamma = 0.4$	$\gamma = 0.6$	$\gamma = 0.8$
RECON	1.037 (0.127)	0.966 (0.112)	0.999 (0.093)	1.0 (0.12)	0.991 (0.091)
RECON+PB	1.002 (0.011)	1.0 (0.019)	1.0 (0.026)	0.995 (0.027)	0.984 (0.023)
RECON*	0.897 (0.165)	1.005 (0.174)	0.986 (0.193)	0.959 (0.177)	0.969 (0.159)
RECON*+PB	1.002 (0.021)	0.997 (0.032)	1.001 (0.036)	0.99 (0.034)	0.978 (0.036)
RCF	1.012 (0.07)	0.991 (0.095)	0.986 (0.153)	0.962 (0.175)	1.01 (0.184)
RCF+PB	0.998 (0.025)	0.995 (0.026)	0.994 (0.035)	0.98 (0.039)	0.973 (0.044)
Random	1.004 (0.035)	1.007 (0.043)	1.003 (0.039)	1.009 (0.034)	1.009 (0.041)

Table 10. Discrimination mean and SD for direct discrimination